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THE EVOLUTION OF CONVENTIONS

BY H. PEYTON YOUNG¹

Consider an n -person game that is played repeatedly, but by different agents. In each period, n players are drawn at random from a large finite population. Each player chooses an optimal strategy based on a sample of information about what others players have done in the past. The sampling defines a stochastic process that, for a large class of games that includes coordination games and common interest games, converges almost surely to a pure strategy Nash equilibrium. Such an equilibrium can be interpreted as the “conventional” way of playing the game. If, in addition, the players sometimes experiment or make mistakes, then society occasionally switches from one convention to another. As the likelihood of mistakes goes to zero, only some conventions (equilibria) have positive probability in the limit. These are known as *stochastically stable equilibria*. They are essentially the same as the risk dominant equilibria in 2×2 games, but for general games the two concepts differ. The stochastically stable equilibria are computed by finding a path of least resistance from every equilibrium to every other, and then finding the equilibrium that has lowest overall resistance. This is a special case of a general theorem on perturbed Markov processes that characterizes their stochastically stable states graph-theoretically.

KEYWORDS: Stochastic stability, equilibrium selection, Markov process, fictitious play.

The individual is foolish but the species is wise.
Edmund Burke

1. INTRODUCTION

A CONVENTION IS A PATTERN of behavior that is customary, expected, and self-enforcing. Everyone conforms, everyone expects others to conform, and everyone wants to conform given that everyone else conforms.² Familiar examples include driving on the right when others drive on the right, going to lunch at noon if others go at noon, accepting dollar bills in payment for goods if others accept them, and so forth. Conventions need not be symmetric. Men conventionally propose to women. Sailboats on the port tack yield the right-of-way to sailboats on the starboard tack. In some regions, tenant farmers customarily get one-third of the harvest and landlords get two-thirds, whereas in other regions the reverse convention holds.³ For each role in such asymmetric interactions there is a customary and expected behavior, and everyone prefers to follow the behavior expected of him provided that others follow the behavior expected of them. Under these circumstances we say that people follow a convention. A convention is an equilibrium that everyone expects. But how do expectations become established when there is more than one equilibrium?

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² See Lewis (1967).

³ See Bardhan (1984) for an account of sharecropping patterns in India.

One explanation is that some equilibria are *a priori* more reasonable than others. A deductive theory of this type has been proposed by Harsanyi and Selten (1988). A second explanation, proposed by Schelling (1960), is that agents focus their attention on one equilibrium because it is more prominent or conspicuous than the others. Yet a third explanation is that, over time, expectations converge on one equilibrium through positive feedback effects. Suppose that a game is played repeatedly, either by the same or different agents. Past plays have a feedback effect on the expectations and behaviors of those playing the game now because people pay attention to precedent. Eventually, one equilibrium becomes entrenched as the conventional one, not because it is inherently prominent or focal, but because the dynamics of the process happen to select it.

This evolutionary explanation for the origin of conventions has been suggested in a variety of papers, but the precise dynamics of the process by which expectations and behaviors evolve has not been clearly spelled out.⁴ In particular it is not clear whether it works. Does the process converge to an equilibrium, and if so, are all equilibria equally likely to be selected? We shall show that the process converges in an asymptotic sense provided that the underlying game has an acyclic best reply structure, and provided there is sufficient stochastic variability in the players' responses. In this case, society is at or close to a Nash equilibrium most of the time. Not all Nash equilibria are equally likely to be selected, however. In fact, typically only *one* Nash equilibrium will be observed with high probability in the long run. Such an equilibrium is said to be *stochastically stable* (Foster and Young (1990)). Building on work of Freidlin and Wentzell (1984) and Kandori, Mailath, and Rob (1993), we shall show how to compute the stochastically stable equilibria by solving a series of shortest path problems in a graph. This is an application of a more general result (proved in the Appendix) that characterizes graph-theoretically the stochastically stable communication classes of a perturbed finite-state Markov process.

2. OUTLINE OF THE MODEL

We consider a fixed n -person game that is played once each period. The players are drawn at random from a large, finite population of individuals. Each player chooses an optimal strategy based on his beliefs about his environment, which he takes to be stationary. He forms his beliefs by looking at what other agents have done in the recent past. Since gathering information is costly, however, each player knows only a small portion of the history, that is, he bases his current actions on a *sample* of plays from recent time periods. We shall also assume that the players occasionally experiment with different strategies, or simply make mistakes.

⁴ See, for example, Lewis (1967), Axelrod (1986), Sugden (1986), Bicchieri (1990), and Warneryd (1990).

The strategies that the agents choose in the current period are recorded and the game is played again in the next period by another random draw of n agents from the fixed population. Each of these agents takes a random sample of previous plays and reacts accordingly. Actions in earlier periods therefore have a feedback effect on actions by agents in later periods. To emphasize that convergence in this model has nothing to do with learning at the individual level, we can assume that after an agent plays the game once he dies and is replaced by a naive agent of the same type (same sample size and same utility function) but no prior information. Thus each time an agent plays he starts afresh and must ask around to find out what is going on. This assumption is not necessary from a mathematical point of view, but it underscores the fact that we are ignoring learning and reputation effects.

The adaptive dynamics described above define a Markov chain whose states are the histories of play truncated to a finite number of periods. It is similar to fictitious play in that agents choose best replies to other agents' past actions. In fictitious play, however, agents base their decisions on the entire history of actions by other agents. Here we assume that agents base their decisions on *limited* information about actions of other agents in the *recent* past, and they do not always optimize. These assumptions seem less fictitious than fictitious play; hence we call this process *adaptive play*.

For general n -person games, adaptive play need not converge to a Nash equilibrium, either pure or mixed, as we shall show below by example. Nevertheless, there is an important class of games for which it does converge. These games have the property that, from any initial choice of strategies, there exists a sequence of best replies that leads to a strict, pure strategy Nash equilibrium. This class includes, but is substantially more general than, coordination games and common interest games. For these *weakly acyclic* games, adaptive play converges with probability one to a pure strategy Nash equilibrium *provided* that the samples are sufficiently incomplete and the players never make mistakes. Incompleteness creates enough stochastic variability to prevent the process from becoming stuck in suboptimal cycles. Finite memory allows past miscoordinations to be forgotten eventually. Once a given equilibrium has been played for as long as anyone can remember, then this equilibrium becomes entrenched as the "conventional" way of playing the game. It is an absorbing state of the process. One cannot say in advance, of course, which equilibrium will become the conventional one, since this depends on the vagaries of the process and on the initial state. What can be said is that some equilibrium will eventually be selected with probability one, and it will *not* be a mixed strategy equilibrium.

If the players occasionally experiment or make mistakes, however, then more can be said. In this case the process has no absorbing states; rather, it has a stationary distribution that describes the relative frequency with which different states are observed in the long run. We shall show that, if the probability of mistakes is small, then this stationary distribution is concentrated around a particular subset of pure strategy Nash equilibria. In fact, typically it puts almost

all the weight on exactly one equilibrium. This *stochastically stable equilibrium* will be observed with probability close to one when the noise is very small.

This concept differs in an important respect from other notions of equilibrium stability, such as evolutionarily stable strategies.⁵ An evolutionary stable strategy is a strategy (or frequency distribution of strategies) that is restored after a small one-time shock to the system. A stochastically stable distribution is a distribution that is restored repeatedly when the evolutionary process is constantly buffeted by small random shocks. This concept was first defined for general evolutionary processes by Foster and Young (1990). Subsequently it was applied to a discrete model of equilibrium selection in a pioneering paper by Kandori, Mailath, and Rob (1993). They consider an evolutionary learning process defined on symmetric 2×2 games. In each period every player plays every other. Successful strategies are adopted with higher probability than unsuccessful ones, and there is a small probability that players make mistakes. Kandori, Mailath, and Rob show that this stochastic process selects the risk dominant Nash equilibrium when the mistake probability is small.⁶ The techniques of analysis in both papers are similar, and build on the theory of perturbed dynamical systems developed by Freidlin and Wentzell (1984).

Using somewhat different methods, Canning (1992) studies a general class of learning models in which agents adapt their behavior to the current state and occasionally make mistakes. He shows that, under certain regularity conditions, the stationary distribution of the perturbed process converges to a stationary distribution of the unperturbed one. In this paper we shall show how to characterize the support of the limiting stationary distribution (the “stochastically stable set”) by solving a series of shortest path problems in a graph. Very often the support consists of a single absorbing state, in which case we obtain a complete characterization of the limiting distribution, and a unique stochastically stable equilibrium. We then apply this result to compute the stochastically stable equilibria of adaptive play. For 2×2 coordination games we show that the risk dominant equilibrium is the unique stochastically stable equilibrium. In coordination games with more than two strategies, the stochastically stable equilibrium may be neither risk dominant nor Pareto optimal, as we show by example. Although there appears to be no simple formula that characterizes the stable equilibria in the general case, they may be computed by a general algorithm that is efficient to implement.

3. ADAPTIVE PLAY

Let Γ be an n -person game in strategic form, and let S_i be the finite set of strategies available to player i . Let N be a finite population of individuals that

⁵ Models of equilibrium selection based on the concept of evolutionarily stable strategy (ESS) include Axelrod (1984), Fudenberg and Maskin (1990), Samuelson and Zhang (1992), Crawford (1991), and Samuelson (1991a). For other models of evolutionary dynamics in games, see Samuelson (1988), Nachbar (1990), and Friedman (1991).

⁶ Models of equilibrium selection based on the concept of stochastic stability include Young and Foster (1991), Fudenberg and Harris (1992), Kandori and Rob (1992), and Samuelson (1991b). For other stochastic selection models see Evans and Honkapohja (1992a, 1992b), and Kirman (1992).

is partitioned into n nonempty classes $C_1, C_2, \dots, C_n$. Each member of C_i is a candidate to play role i in the game. For example, C_1 is the class of men, C_2 is the class of women, and the game is Battle of the Sexes. We shall assume that all individuals in class i have the same utility function $u_i(s)$ for strategy-tuples $s = (s_1, s_2, \dots, s_n) \in \Pi S_i$, which we shall identify with outcomes.

Let $t = 1, 2, \dots$, denote successive time periods. The game G is played once each period. In period t , one individual is drawn at random from each of the n classes and is assigned to play the appropriate role in the game. It will be convenient to refer to the individual playing role i as player “ i ” even though the identity of this individual may change from one period to the next. Player i chooses a pure strategy $s_i(t)$ from his strategy space according to a rule that will be defined below. The strategy-tuple $s(t) = (s_1(t), s_2(t), \dots, s_n(t))$ is recorded and will be referred to as the *play at time t* . The *history* of plays up to time t is the sequence $h(t) = (s(1), s(2), \dots, s(t))$. We assume that the histories are anonymous: it does not matter *who* played a given strategy in a given period, only that it was played by someone.

The players decide how to choose their strategies as follows. Fix integers k and m such that $1 \leq k \leq m$. In period $t + 1$ ($t \geq m$) each player inspects k plays drawn without replacement from the most recent m periods $t, t - 1, t - 2, \dots, t - m + 1$. The draws are independent for the various players. One way to think about the sampling procedure is that each player “asks around” to find out how the game was played in recent periods. He stops when he has learned about k different plays within the last m periods (say because he has reached his capacity to retain information). Another way of thinking about the sampling procedure is that each agent passively hears about certain precedents, and k is the number of precedents that come to the agent’s attention. In either case, the fraction k/m measures the *completeness* of the agents’ information relative to the surviving precedents. It is not necessary to assume that every subset of k precedents out of the last m is *equally likely* to constitute an agent’s information. For example, an agent might be more likely to hear about recent precedents than more dated ones. It is enough to assume that every subset of k has a positive probability of being agent i ’s information for every i .⁷

Assume for the sake of generality that the first m plays are randomly selected. Thus we can think of the sampling process as beginning in period $t = m + 1$ from some arbitrary initial sequence of m plays $h(m) = (s(1), s(2), \dots, s(m))$. We then obtain a finite Markov chain on the state space H consisting of all sequences of length m drawn from ΠS_i , beginning with some arbitrary “initial” state $h(m)$.

SUCCESSOR: A *successor* of a state $h \in H$ is any state $h' \in H$ obtained by deleting the left-most element of h and adjoining a new right-most element.

The process moves from the current state h to a successor state h' in each period according to the following transition rule. For each $s \in S_i$, let $p_i(s|h)$ be

⁷ The model can be modified by assuming that different agents have different sample sizes. An application of this type is described in Young (1993).

the probability that agent i chooses s . We assume that $p_i(\cdot)$ is a *best-reply distribution* in the sense that $p_i(s|h) > 0$ if and only if there exists a sample of size k to which s is i 's best reply, and that $p_i(s|h)$ is independent of t . If s is the right-most element of h , the probability of moving from h to h' is

$$(1) \quad P_{hh'}^0 = \prod_{i=1,n} p_i(s_i|h).$$

$P_{hh'}^0 = 0$ if h' is not a successor of h . We call the process P^0 *adaptive play with memory m and sample size k* .

4. CONVERGENCE OF ADAPTIVE PLAY WHEN THERE ARE NO MISTAKES

Let us begin by observing that h is an absorbing state of this process if and only if it consists of a strict pure strategy Nash equilibrium played m times in succession. Suppose, indeed, that $h = (s^1, \dots, s^m)$ is an absorbing state. For each agent i let s_i be i 's best reply to some subset of k plays drawn from h , and let $s = (s_1, \dots, s_n)$. By assumption, there is a positive probability of moving from h to $h' = (s^2, \dots, s^m, s)$ in one period. Since h is absorbing, $h = h'$ and hence $s^1 = s^2$. Continuing in this fashion we conclude that $s^1 = s^2 = \dots = s^m = s$. Hence $h = (s, s, \dots, s)$. By construction, s_i is a best reply to some sample of k elements from h . Hence s_i is a best reply to s_{-i} for each i . It must also be a unique best reply to s_{-i} , because otherwise the process could move to a successor that is different from h . So s is a strict, pure strategy Nash equilibrium. Conversely, any state h consisting of m repetitions of a strict, pure strategy Nash equilibrium is clearly an absorbing state. Such a state will be called a *convention*.

If adaptive play converges to an absorbing state, then clearly the game must have a strict Nash equilibrium in pure strategies. The existence of such an equilibrium is not a sufficient condition for convergence, however. Consider the following variation of an example due to Shapley:

EXAMPLE 1:

	a	b	c	d
a	2, 1	0, 0	1, 2	-1, -1
b	1, 2	2, 1	0, 0	-1, -1
c	0, 0	1, 2	2, 1	-1, -1
d	-1, -1	-1, -1	-1, -1	3, 3

Here d is a best response to itself, but it is not a best response to any mixture of a , b , and c . If the initial state does not involve d , then adaptive play (like fictitious play) cycles. Consider, for example, the case where $m = 2$ and $k = 1$. Let the first two plays be ($\overset{a}{a}$) and ($\overset{c}{c}$). In period 3, Column will sample one of Row's previous two choices (both a) and react by playing c . Row will sample one of Column's previous two choices (a or c) with equal probability and react by playing a or c . So the next play will be ($\overset{a}{c}$) or ($\overset{c}{c}$) with equal probability. The

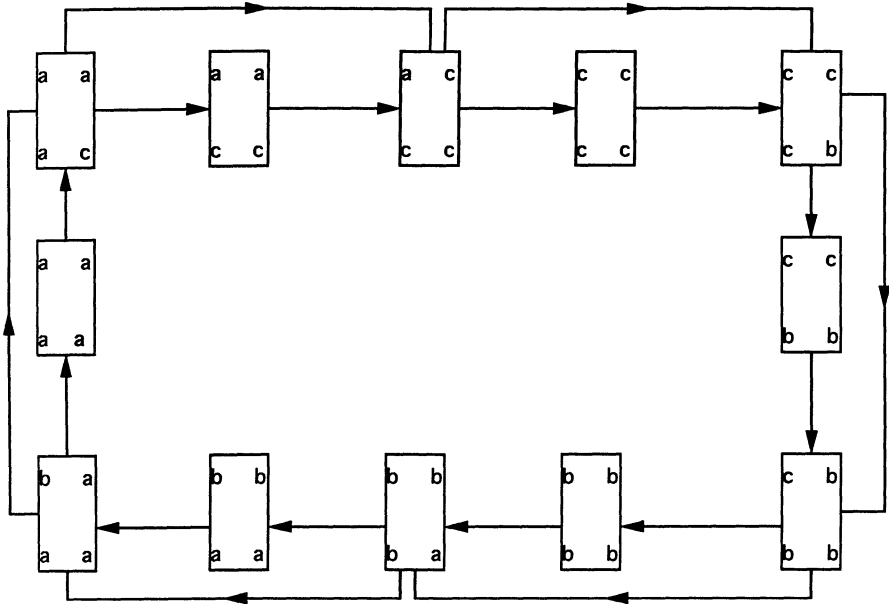


FIGURE 1.—A recurrent communication class of adaptive play with $m = 2$, $k = 1$.

process therefore moves from state $\begin{bmatrix} a & a \\ a & c \end{bmatrix}$ to state $\begin{bmatrix} a & a \\ c & c \end{bmatrix}$ with probability one-half, and to state $\begin{bmatrix} c & c \\ c & c \end{bmatrix}$ with probability one-half. The subsequent transitions form a cycle of length six imbedded within a cycle of length twelve, as shown in Figure 1. This cycle constitutes a recurrent communication class of the Markov process defined by (1).

When cycling is built into the best reply structure of the game, as in the above example, we cannot expect adaptive play to converge. Nevertheless, there are many games that do not have a cyclic best-reply structure. Consider a two-person coordination game in which both agents have the same number of strategies, and each agent strictly prefers to play his j th strategy if and only if the other agent plays his j th strategy for every j . Clearly there is no cycling problem here: once one of them chooses a pure strategy and the other responds optimally, then they have achieved a coordination equilibrium.

To take another example, suppose that the agents have *common interests*: for every two strategy tuples s and s' , either everyone prefers s to s' or everyone prefers s' to s . Assume further that no two strategy-tuples are payoff equivalent. Given an arbitrary strategy-tuple s that is not a Nash equilibrium, there exists some agent i who can do better by playing s'_i instead of s_i . Let $s' = (s'_i, s_{-i})$. If s' is not Nash equilibrium, there is some agent j who can do better by playing s''_j instead of s'_j . Let $s'' = (s''_j, s'_{-j})$, and so forth. At each step of this adjustment process everyone's utility increases, so the process cannot cycle and it must end at a pure strategy Nash equilibrium. Moreover, this equilibrium

must be strict because of the assumption that no two strategy-tuples are payoff equivalent and the players have common interests.

This construction can be generalized as follows. Let Γ be an n -person game in normal form on a finite strategy space ΠS_i . Define the *best-reply graph* of Γ as follows: each vertex is an n -tuple of strategies $s \in \Pi S_i$, and for every two vertices s and s' there is a directed edge $s \rightarrow s'$ if and only if $s \neq s'$ and there exists exactly one agent i such that s'_i is a best reply to s_{-i} and $s'_{-i} = s_{-i}$.

ACYCLIC GAME: A game Γ is *acyclic* if its best reply graph contains no directed cycles. It is *weakly acyclic* if, from any initial vertex s , there exists a directed path to some vertex s^* from which there is no exiting edge (a *sink*).

Every sink of the best reply graph is clearly a strict Nash equilibrium in pure strategies. So a game is weakly acyclic if, and only if, from every strategy-tuple there exists a finite sequence of best replies by one agent at a time that ends in a strict, pure strategy Nash equilibrium. We shall show that, for this class of games, adaptive play converges with probability one provided that sampling is sufficiently incomplete and the players do not make mistakes.

Let Γ be a weakly acyclic n -person game. For each strategy-tuple s , let $L(s)$ be the length of a shortest directed path in the best reply graph from s to a strict Nash equilibrium, and let $L_\Gamma = \max_s L(s)$.

THEOREM 1: *Let Γ be a weakly acyclic n -person game. If $k \leq m/(L_\Gamma + 2)$, then adaptive play converges almost surely to a convention.*

PROOF: Fix k and m , where $k \leq m/(L_\Gamma + 2)$. We shall show that there exists a positive integer M , and a positive probability p , such that from any state h , the probability is at least p that adaptive play converges within M periods to a convention. M and p are time-independent and state-independent. Hence the probability of *not* reaching a convention after at least rM periods is at most $(1 - p)^r$, which goes to zero as $r \rightarrow \infty$.

Let $h = (s(t - m + 1), \dots, s(t))$ be the state in period $t \geq m$. In period $t + 1$ there is a positive probability that each of the n agents samples the last k plays in h , namely, $(s(t - k + 1), \dots, s(t)) = \eta$. There is also a positive probability that, from periods $t + 1$ to $t + k$ inclusive, every agent draws the sample η every time. Finally, there is a positive probability that, if an agent has a choice of several best replies to η , then he will choose the same one k times in succession. Thus there is a positive probability of a run $(s, s, \dots, s)$ from periods $t + 1$ to $t + k$ inclusive. Note that this argument depends on the agents' memory being at least $2k - 1$, since otherwise they could not choose the sample η in period $t + k$.

Suppose that s happens to be a strict Nash equilibrium. There is a positive probability that, from periods $t + k + 1$ through $t + m$, each agent will sample only the last k plays, in which case the unique best response of each agent i is

s_i . So they play s for $m - k$ more periods. At this point an absorbing state has been reached, and they continue to play s forever.

Suppose instead that s is not a strict Nash equilibrium. Since Γ is weakly acyclic, there exists a directed path $s, s', \dots, s^r$ in the best reply graph such that s^r is a strict Nash equilibrium. The first edge on this path is $s \rightarrow s'$. Let i be the index such that $s'_{-i} = s_{-i}$ and s'_i is a best reply to s_{-i} . Consider the event in which agent i samples from the run of s established in periods $t + 1$ to $t + k$ and responds by playing s'_i , while every agent $j \neq i$ draws the sample $\eta = (s(t - k + 1), \dots, s(t))$. By assumption, a best response of every agent j to this sample is s_j . These events occur together with positive probability, and there is a positive probability that they occur in every period from $t + k + 1$ to $t + 2k$, assuming that $m \geq 3k - 1$. The result is a run of $s' = (s'_i, s_{-i})$ for k periods in succession.

Continuing in this fashion, we see that there is a positive probability of obtaining a run of s , followed by a run of s' ... followed eventually by a run of s^r . Each run is of length k , and the run of s^r occurs from period $t + kr + 1$ to $t + kr + k$. To reach this point may require that some agent look back $kr + 2k - 1$ periods, namely, from period $t + kr + k$ to period $t - k + 1$. This is possible because of the assumption that $k \leq m/(L_\Gamma + 2)$.

After this, the process can converge to the absorbing state $(s^r, s^r, \dots, s^r)$ by period $t + kr + m$ if each agent samples the previous k plays from periods $t + kr + k + 1$ to $t + kr + m$ inclusive.

Since $r \leq L_\Gamma$, we have established that, given an initial state h , there is a probability $p_h > 0$ of converging to an absorbing state within $M = kL_\Gamma + m$ periods. Letting $p = \min_{h \in H} p_h > 0$, it follows that from any initial state the process converges with probability at least p to an absorbing state within at most M periods. This completes the proof.

We do not claim that the bound $k \leq m/(L_\Gamma + 2)$ is best possible for all weakly acyclic games. Without incomplete sampling, however, the process may not converge to a pure Nash equilibrium. Consider the following version of "Battle of the Sexes":

EXAMPLE 2:

	Yield	Not Yield
Yield	0, 0	1, $\sqrt{2}$
Not Yield	$\sqrt{2}$, 1	0, 0

Let $k = m$, so that both players sample the same m plays in each period. Consider any initial sequence of m plays in which the players have always miscoordinated, that is, they both yielded or they both failed to yield in each period. Let f be the frequency with which they yielded in this sequence. In the next period, Row yields if and only if $1 - f > f\sqrt{2}$, and Column does the same. So they miscoordinate again. (Since f is rational, the inequality is always strict

so we never have to consider ties.) Thus, if they begin in a state of perfect miscoordination, then they miscoordinate forever.

The same holds if memory is unbounded, as in fictitious play: if they miscoordinate on the first move, then they will continue to miscoordinate forever. In this case the frequency distributions converge to the mixed Nash equilibrium of the game in which both players yield with probability $1/(1 + \sqrt{2})$. Note, however, that their *behaviors* do not converge to the behavior specified by the mixed Nash equilibrium, because in each period their moves are correlated: either both yield or both do not yield. So even in 2×2 games fictitious play need not converge in a behavioral sense. This problem does not arise in adaptive play, because when it converges to an absorbing state, this state must correspond to a Nash equilibrium in pure strategies. Hence the behaviors also converge.

The feature of adaptive play that allows it to break out of suboptimal cycles is incomplete sampling, which introduces stochastic variation into the players' responses. There is a possibility that they will coordinate by chance, and if they do so often enough the process eventually locks in to a pure strategy Nash equilibrium. This equilibrium then becomes the conventional way of playing the game, because for as long as anyone can remember, the game has always been played in this way. Therefore sampling does not matter any more, because no matter what samples the agents take, their optimal responses will be to play the equilibrium that is already in place.

5. ADAPTIVE PLAY WITH MISTAKES

Theorem 1 relies on the assumption that, while agents base their decisions on limited information, they always choose a best response given their information. This assumption is clearly unrealistic. Agents sometimes make mistakes; they may also experiment with nonoptimal responses. In this case the stochastic process does not converge to an absorbing state, because it has no absorbing states. Mistakes constantly perturb the process away from equilibrium. If we assume, however, that all mistakes are possible and that the mistake probabilities are time-independent, then the process does have a unique stationary distribution. Hence we can study its asymptotic behavior. When the probability of mistakes is small, we shall show that this stationary distribution is concentrated around a particular convention (or, in the event of ties, a subset of conventions). These are the *stochastically stable* conventions—the ones that will be observed with positive probability in the long run when the noise is small but nonvanishing. We therefore obtain a theory of equilibrium selection. As we shall see below, it yields (except in the 2×2 case) quite different answers than the theory of Harsanyi and Selten (1988).

Our model of mistakes generalizes an approach pioneered by Canning (1992) and Kandori, Mailath, and Rob (1993). Fix the sample size k and memory m . Suppose that, in each time period, there is some small probability $\varepsilon \lambda_i > 0$ that player i experiments by choosing a strategy randomly from S_i instead of

optimizing based on a sample of size k . The ratio λ_i/λ_j is the relative probability with which a player of type i experiments as compared to a player of type j . The factor ε determines the probability with which players in general experiment. The event that i experiments is assumed to be independent of the event that j experiments for every $i \neq j$. For every i , let $q_i(s|h)$ be the conditional probability that i chooses $s \in S_i$ given that i experiments and the process is in state h , where $\sum_{s \in S_i} q_i(s|h) = 1$ for every i and h . We shall assume that $q_i(s|h)$ is independent of t , and that $q_i(s|h) > 0$ for all $s \in S_i$. The latter assumption is made for ease of exposition; similar results hold provided the $q_i(\cdot)$ have enough support that every state is reachable from every other state in a finite number of periods by agents who experiment.

A priori we do not know the distributions $\mathbf{q} = (q_1(\cdot), q_2(\cdot), \dots, q_n(\cdot))$ or the relative probabilities of experimentation $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n)$. It turns out, however, that this does not matter. If the overall probability of experimentation ε is small, and if the agents experiment independently of one another, then the selected equilibria are *independent* of $\mathbf{q}$ and λ .

The perturbed process may be described as follows. Suppose that the process is in state h at time t . Let J be a subset of j players, $1 \leq j \leq n$. The probability is $\varepsilon^j (\prod_{j \in J} \lambda_j) (\prod_{j \notin J} (1 - \varepsilon \lambda_j))$ that exactly the players in J experiment and the others do not. Conditional on this event, the transition probability of moving from h to h' is

$$Q_{hh'}^J = \prod_{j \in J} q_j(s_j|h) \prod_{j \notin J} p_j(s_j|h) \quad \text{if } h' \text{ is a successor of } h \text{ and } s \text{ is the right-most element of } h';$$

$$Q_{hh'}^J = 0 \quad \text{if } h' \text{ is not a successor of } h.$$

If no agent experiments, then the transition probability of moving from h to h' in one period is $P_{hh'}^0$ as defined in (1). This event has probability $\prod_{i=1,n} (1 - \varepsilon \lambda_i)$. The perturbed Markov process therefore has the transition function:

$$(2) \quad P_{hh'}^\varepsilon = \left(\prod_{i=1,n} (1 - \varepsilon \lambda_i) \right) P_{hh'}^0 + \sum_{J \subseteq N, J \neq \emptyset} \varepsilon^{|J|} \left(\prod_{j \in J} \lambda_j \right) \left(\prod_{j \notin J} (1 - \varepsilon \lambda_j) \right) Q_{hh'}^J.$$

The process P^ε will be called *adaptive play with memory m , sample size k , experimentation probabilities $\varepsilon \lambda_i$ and experimentation distributions q_i* . Note that P^0 is the process defined in (1), which we shall refer to as the *unperturbed* process.

6. ASYMPTOTIC BEHAVIOR OF ADAPTIVE PLAY

We shall now characterize the asymptotic behavior of process (2) when the overall probability of experimenting ε is close to zero. Let h and h' be two distinct states. If P^ε is in state h at time t , there is a positive probability that all players will experiment for m periods in succession. Thus there is a positive probability that the process arrives at state h' at time $t + m$, so P^ε is irreducible. It is aperiodic because the process can move from h to h in exactly m

periods, and also in exactly $m + 1$ periods. Hence P^ε has a unique stationary distribution μ^ε satisfying the equation $\mu^\varepsilon P^\varepsilon = \mu^\varepsilon$. The process is strongly ergodic, and (with probability one) μ_h^ε is the cumulative relative frequency with which state h will be observed when the process runs for a very long time. It is also the probability that h will be observed at any given time t , provided that t is sufficiently large.

The following concept was introduced by Foster and Young (1990).

STOCHASTIC STABILITY: A state $h \in H$ is *stochastically stable* relative to the process P^ε if $\lim_{\varepsilon \rightarrow 0} \mu_h^\varepsilon > 0$.

Over the long run, states that are not stochastically stable will be observed infrequently compared to states that are, provided that the probability of mistakes ε is small. If there is a *unique* stochastically stable state, then it will be observed almost all of the time when ε is small.

We shall now show how to compute the stochastically stable states of adaptive play for a general n -person game. In the process we shall show that these states are essentially *independent* of the particular mistake distributions q_i and the mistake probabilities λ_i . Then we shall specialize to the case of weakly acyclic games, and show that if $k \leq m/(L_T + 2)$, every stochastically stable state is a convention, and if k and m are sufficiently large, then typically it is unique.

MISTAKE: Let h' be a successor of h and let s be the right-most element of h' . A *mistake* in the transition $h \rightarrow h'$ is a component s_i of s that is not an optimal response by agent i to any sample of size k from h .

A mistake can only arise if a player experiments, but an experimental choice need not be a mistake, since it could (by chance) be an optimal choice.

RESISTANCE: For any two states h, h' the *resistance* $r(h, h')$ is the total number of mistakes involved in the transition $h \rightarrow h'$ if h' is a successor of h ; otherwise $r(h, h') = \infty$.

Let us now view the state space H as the vertices of a directed graph. For every pair of states h, h' insert a directed edge $h \rightarrow h'$ if $r(h, h')$ is finite, and let $r(h, h')$ be its "weight" or "resistance." The edges of zero resistance correspond to the transitions that occur with positive probability under P^0 . Let $H_1, H_2, \dots, H_J$ be the *recurrent communication classes* of P^ε . These classes are disjoint, and they are characterized by the following three properties: (i) from every state there is a path of zero resistance to at least one of the classes H_i ; (ii) within each class H_i there is a path of zero resistance from every state to every other; (iii) every edge exiting from H_i has positive resistance.

Given any two distinct classes H_i and H_j consider all directed paths that begin in H_i and end in H_j . There is at least one such path, because the perturbed process P^ε is irreducible. Among all such paths, find one with *least*

total resistance, and let this resistance be denoted by r_{ij} . Clearly $r_{ij} \geq 0$. Computing r_{ij} amounts to solving a shortest path problem in a directed graph, for which there exist very efficient algorithms. Note that r_{ij} is independent of which vertex in H_i is the starting point and which vertex in H_j is the termination point, because every two states *within* the same class are accessible from each other by paths of zero resistance.

Now define a new directed graph $\mathcal{G}$ as follows: there is one vertex i for each recurrent communication class H_i , and for every distinct $1 \leq i, j \leq J$ the directed edge (i, j) has “weight” or *resistance* r_{ij} . The following concept is due to Friedlin and Wentzell (1984).

i-TREE: An *i*-tree in $\mathcal{G}$ is a spanning tree such that from every vertex $j \neq i$ there is a unique path directed from j to i .

For every vertex i let $\mathcal{T}_i$ be the set of all *i*-trees on $\mathcal{G}$. The *resistance* of an *i*-tree $\tau \in \mathcal{T}_i$ is the sum of the resistances of its edges,

$$(3) \quad r(\tau) = \sum_{(i,j) \in \tau} r_{ij}.$$

STOCHASTIC POTENTIAL: The *stochastic potential* of the recurrent class H_i is the least resistance among all *i*-trees:

$$(4) \quad \gamma_i = \min_{\tau \in \mathcal{T}_i} r(\tau).$$

Computing γ_i for a given set of weights r_{ij} is a standard problem in combinatorial optimization known as the *arborescence problem*. There exist algorithms for solving it in the order of $|J|^2$ steps (see Chu and Liu (1965), Edmonds (1967), and Tarjan (1977)). Since there are $|J|$ vertices in G , and one arborescence problem must be solved for each, the potential function can be computed in $O(|J|^3)$ steps.

Note that the numbers r_{ij} depend only on the *number* of mistakes in making various transitions, not on the relative probability with which specific mistakes are made. Hence the potential function is independent of the parameters λ_i and q_i . This is important, for in applications one would rarely know the relative probabilities of various mistakes, only that they are possible. What matters is that the probability of mistakes is small, that all mistakes are possible, and the agents make them independently of one another.

THEOREM 2: Let Γ be an n -person game on a finite strategy space. The stochastically stable states of adaptive play P^ε are the states contained in the recurrent communication classes of P^0 with minimum stochastic potential. These states are independent of the experimentation probabilities λ_i and the experimentation distributions q_i so long as they have full support.

COROLLARY: *If Γ is weakly acyclic and $k \leq m/(L_\Gamma + 2)$, the stochastically stable states of adaptive play are the convention(s) of minimum stochastic potential.*

Theorem 2 follows from a general theorem on perturbed Markov processes that we prove in the Appendix. The corollary is a direct consequence of Theorems 1 and 2.

The stochastically stable states are computed in three steps. First we identify the recurrent communication classes of the process P^0 without mistakes. For a general n -person game these classes can be quite complex. If the game is weakly acyclic and the sampling is sufficiently incomplete, however, then Theorem 1 tells us that the recurrent classes correspond one-to-one with the strict pure strategy Nash equilibria, which are easy to identify. The second step is to compute the least resistance in moving from every recurrent class to every other. In theory this involves solving a series of shortest path problems, but in practice the computation can often be made directly from the payoff matrix of the game. The third and final step is to construct a complete directed graph with these resistances as weights, and to find the arborescence having least weight. This identifies the stochastically stable convention(s), which is unique except in the event of ties. In the remainder of the paper we shall illustrate the technique for 2×2 and 3×3 matrix games, and show how the stochastically stable equilibria relate to standard concepts of equilibrium selection such as risk dominance.

7. THE 2×2 CASE

Let Γ be a 2×2 matrix game with two strict Nash equilibria in pure strategies. It is clear that Γ is acyclic and $L_\Gamma = 1$. Without loss of generality we may write Γ in the form

	1	2
1	a_{11}, b_{11}	a_{12}, b_{12}
2	a_{21}, b_{21}	a_{22}, b_{22}

where $a_{11} > a_{21}$, $b_{11} > b_{12}$, $a_{22} > a_{12}$, and $b_{22} > b_{21}$. The strict, pure strategy Nash equilibria are (1, 1) and (2, 2). Theorem 1 implies that, if $k \leq m/3$, adaptive play without mistakes has two absorbing states: $h_1 = ((1, 1), (1, 1), \dots, (1, 1))$ and $h_2 = ((2, 2), (2, 2), \dots, (2, 2))$. To determine which of these states is stochastically stable, we must compute the path of least resistance from h_1 to h_2 , and the path of least resistance from h_2 to h_1 .

Let h_1 be the state at time $t = m$. To go from h_1 to h_2 requires that at least one player choose strategy 2 by mistake. Moreover, he must choose strategy 2 so often that the other's optimal reply is also strategy 2 for at least one sample of size k , for otherwise the process cannot lock in to the absorbing state h_2 .

TABLE 1

A SUCCESSION OF k' MISTAKES BY ROW CAUSES THE PROCESS TO CONVERGE TO h_2 .
 2^* DENOTES A MISTAKEN CHOICE OF 2, 1(2) AN OPTIMAL CHOICE OF EITHER 1 OR 2.

Period	1	2... m	$m+1$	$m+2$	...	$m+k'$	$m+k'+1$	...	$m+k'+k$	$m+k'+k+1...$
Row	1	1...1	2*	2*		2*	1	...	1(2)	2
Column	1	1...1	1	1		1	2		2	2

Suppose, for example, that Row chooses 2 by mistake from periods $t = m + 1$ to $t = m + k'$ inclusive, where $k' \leq k$. From then on Row makes no further mistakes. These choices are marked 2^* in Table I.

If Column draws a sample that includes these k' choices of 2, as well as $k - k'$ choices of 1, then Column's best reply is 2 provided that

$$(1 - k'/k)b_{12} + (k'/k)b_{22} \geq (1 - k'/k)b_{11} + (k'/k)b_{21};$$

that is,

$$(5) \quad k' \geq \frac{b_{11} - b_{12}}{b_{11} - b_{12} - b_{21} + b_{22}} k.$$

If equality holds in (5) then strategy 2 is among Column's best replies, so Column will play it with positive probability.

Suppose that (5) holds and that Column's sample happens to include Row's mistakes in every period from $m + k' + 1$ to $m + k' + k$ inclusive. This event has positive probability provided that m is sufficiently large relative to k . (It suffices that $m \geq 2k$.) Then Column's best reply is to play 2 from periods $m + k' + 1$ to $m + k' + k$ and none of these choices are mistakes. In period $m + k' + k + 1$ Row may sample Column's choices of 2, while Column samples Row's choices of 2, in which case their best replies are to play 2. In the next period there is again a positive probability that both sample enough choices of 2 to want to play 2 again, and so forth. So with positive probability the process converges to the absorbing state h_2 with no further mistakes. In other words, k' mistakes are sufficient to move the process from h_1 to h_2 provided that k' satisfies (5) and m/k is large enough.

Similarly, the process converges with positive probability to h_2 if Column chooses 2 by mistake k'' times, where

$$k'' \geq \frac{a_{11} - a_{21}}{a_{11} - a_{12} - a_{21} + a_{22}} k.$$

Let

$$R_1 = \min \left\{ \frac{a_{11} - a_{21}}{a_{11} - a_{12} - a_{21} + a_{22}}, \frac{b_{11} - b_{12}}{b_{11} - b_{12} - b_{21} + b_{22}} \right\}.$$

For every real number x , let $[x]$ denote the least integer greater than or equal to x . We have just shown that the resistance in going from h_1 to h_2 is $[R_1 k]$. A similar argument shows that the resistance in going from h_2 to h_1 is $[R_2 k]$ where

$$R_2 = \min \left\{ \frac{a_{22} - a_{12}}{a_{11} - a_{12} - a_{21} + a_{22}}, \frac{b_{22} - b_{21}}{b_{11} - b_{12} - b_{21} + b_{22}} \right\}.$$

If $R_1 > R_2$, then $(1, 1)$ *risk dominates* $(2, 2)$ in the terminology of Harsanyi and Selten (1988). The pair $(1, 1)$ *weakly risk dominates* $(2, 2)$ if $R_1 \geq R_2$. If $R_1 > R_2$, then the unique stochastically stable convention is h_1 for all sufficiently large values of k and m/k . If $R_1 = R_2$, then both h_1 and h_2 are stochastically stable conventions for all sufficiently large values of k and m/k .

Note that the discrimination of the process grows with the sample size, because the players can only respond to frequency distributions that involve integers between 0 and k . For all sufficiently large k , the process can discriminate any difference in resistance between the two equilibria. This leads to the following definition.

GENERIC STABILITY: A strict pure strategy Nash equilibrium is *generically stable* if the associated convention is stochastically stable for all sufficiently large k and m such that $k \leq m/(L_T + 2)$.

THEOREM 3: Let Γ be a 2×2 matrix game with two strict Nash equilibria in pure strategies. The generically stable equilibria are the weakly risk dominant Nash equilibria.

Kandori, Mailath, and Rob (1991) obtain a similar result for symmetric games using a somewhat different dynamic adjustment process. In their model, there is a single homogeneous population of N individuals who play a symmetric 2×2 game. At each time period t , z_t is the current number of players who have “adopted” strategy 1, and $N - z_t$ is the number who have adopted strategy 2. For $i = 1, 2$ let $\pi_i(z_t)$ be the total payoff to strategy i in state z_t when every player is matched once against every other. The dynamical assumption is that players adopt successful strategies more often than unsuccessful strategies. That is, there is a deterministic dynamic $z_{t+1} = f(z_t)$ such that

$$\text{for all } 0 \leq z_t < N, \quad z_{t+1} \geq z_t \quad \text{if and only if} \quad \pi_1(z_t) \geq \pi_2(z_t).$$

In addition, there is a small probability ε that each player will switch from playing strategy 1 to strategy 2 or vice versa, where the switches are independent across players. Using similar techniques to those developed here, Kandori, Mailath, and Rob show that the resulting irreducible, aperiodic process $P(N, \varepsilon)$

has a stationary distribution that, for all sufficiently large N , puts all of the probability on the risk dominant equilibria as ε goes to zero.

8. THE 3×3 CASE

When the agents have three or more strategies, there is no simple formula analogous to risk dominance that identifies the stochastically stable equilibria. First, the path of least resistance must be computed from every equilibrium to every other. Then a minimum arborescence problem must be solved for each equilibrium. We shall illustrate by solving an example.

EXAMPLE 3:

	1	2	3
1	6, 6	0, 5	0, 0
2	5, 0	7, 7	5, 5
3	0, 0	5, 5	8, 8

The pairs (i, i) are the strict, pure strategy Nash equilibria, $i = 1, 2, 3$. Let h_i denote the convention in which (i, i) is played m times in succession. Theorem 1 says that these are the absorbing states of the unperturbed process provided that $k \leq m/3$. Let us compute the path of least resistance from every convention to every other.

Suppose that the perturbed process is in state h_1 . To exit to h_2 or h_3 , one agent must choose a sufficient number of 2's or 3's (or both) to cause the other agent to choose 2 or 3. Since the game is symmetric, it does not matter which player makes the mistakes and which player reacts. Assume that the Column player chooses 2 at least $k'' = [(1/8)k]$ times in succession. If Row samples these choices (plus $k - k''$ choices of strategy 1), then Row's best reply is also strategy 2. At this point there is a positive probability that the process will converge to h_2 with no further mistakes. (This relies on the assumption that $k \leq m/3$.) It may be checked that this is the least number of mistakes to go from h_1 to h_2 by any route. Thus the least resistant path is *direct* in the sense that it only involves strategies 1 and 2.

Not all paths of least resistance are direct, however. For example, suppose that the process is in state h_3 and we want to exit to state h_2 . The direct route is for one player to choose strategy 2 by mistake at least $[(3/5)k]$ times, which causes the other player to reply with strategy 2. But if one player chooses strategy 1 by mistake at least $[(3/8)k]$ times (and at most $(5/6)k$ times), then the best reply of the other player is strategy 2. Thus the indirect route has lower resistance when k is large enough.

Another example of an indirect route involves the transition from h_1 to h_3 . The direct route requires some player to make $[(5/8)k]$ mistaken choices of strategy 3, since otherwise strategy 1 or 2 is a better reply. But there is a path of less resistance, namely, go first to h_2 and then to h_3 . The total resistance of this

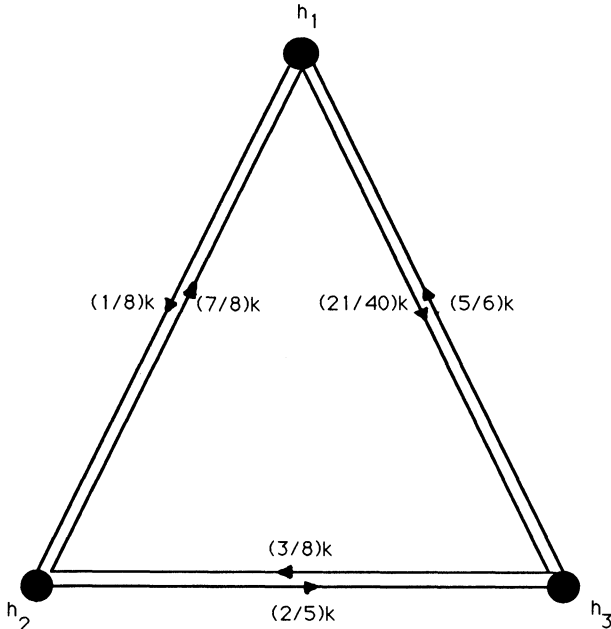


FIGURE 2.—Pairwise resistances (unrounded) for the pure strategy equilibria of Example 3.

path is $[(1/8)k] + [(2/5)k]$, which is approximately $(21/40)k$ when k is large. The resistances between every pair of equilibria are shown in Figure 2.

For each vertex h_i there are three h_i -trees, and the h_i -tree of least resistance determines the stochastic potential of h_i , as shown in Figure 3.

It is readily seen that the least resistant tree is rooted at h_2 and has total resistance $[(1/8)k] + [(3/8)k]$. For all sufficiently large k this is the unique tree of least resistance, so h_2 is the unique generically stable convention. The risk dominant equilibrium, however, is strategy 3. It is also the Pareto optimal equilibrium. To verify risk dominance, one checks that strategy 3 risk dominates strategy 2 in the subgame restricted to strategies 2 and 3, and that strategy 3 risk dominates strategy 1 in the subgame restricted to strategies 1 and 3. (See Harsanyi and Selten (1988).)

There are essentially two respects in which risk dominance and stochastic stability differ. First, they employ different notions of resistance. Strategy i risk dominates j if it requires fewer mistakes to go from j to i than from i to j within the subgame consisting of just these two strategies (i.e., by a direct route). Stochastic stability requires us to look at all transitions from i to j , including those that involve other strategies. The second distinction is that risk dominance is only defined when there is one strategy that risk dominates *every* other in pairwise comparisons. Such a strategy may not exist because of cycles in

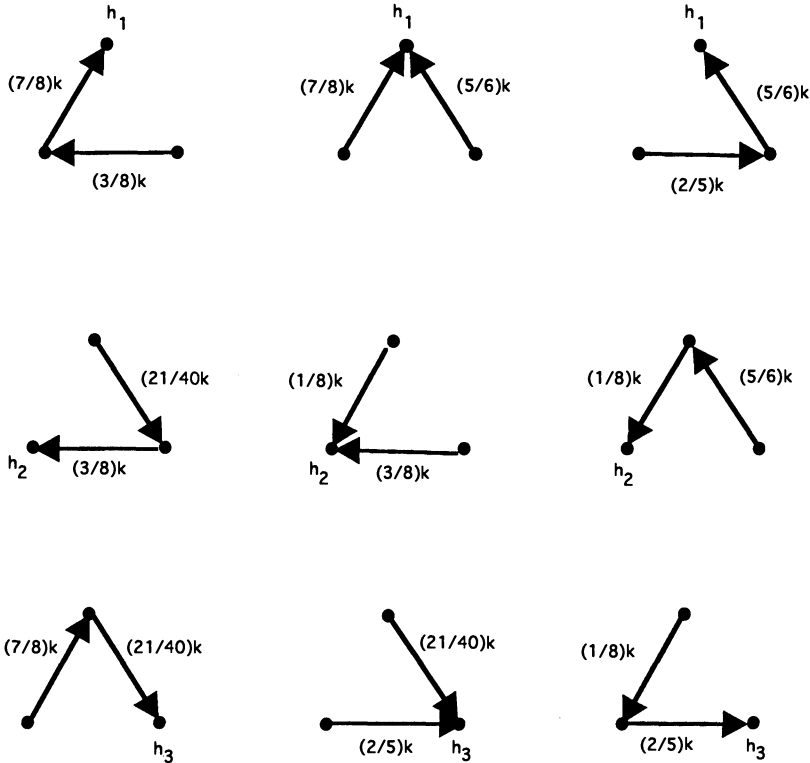


FIGURE 3.—The nine h_i -trees ($i = 1, 2, 3$). Resistances are not rounded.

the pairwise comparisons. Stochastic stability, by contrast, relies on a global criterion (the resistance of spanning trees) to compare the stability of different strategies. The difference between the two concepts may therefore be summarized as follows: risk dominance selects the equilibrium that is easiest to flow into from every other equilibrium considered in isolation (assuming such an equilibrium exists). Stochastic stability selects the equilibrium that is easiest to flow into from all other states combined, including both equilibrium and nonequilibrium states.

9. CONCLUSION

In this paper we have shown how an equilibrium can evolve in a game that is played repeatedly by different agents. The model is similar to fictitious play in that agents' expectations are shaped by precedent. It differs in that agents base their choices on an incomplete knowledge of recent precedents and they occasionally make mistakes. These assumptions seem more natural than the

deterministic dynamics of fictitious play, so we can justify them on the grounds of realism. They also play an important technical role: by introducing noise into the dynamic adjustment process, they select among pure strategy Nash equilibria for weakly acyclic games, and among more complex regimes for general n -person games. Unlike other evolutionary models, these perturbations are not one-shot affairs but form an integral part of the dynamics. By incorporating noise directly into the dynamics, one is led to a different criterion of equilibrium selection than classical notions like evolutionary stability and risk dominance.

Several questions remain to be explored. One is the sensitivity of the equilibrium selected to the way in which the model is specified. We have shown that the stochastically stable equilibria are invariant with respect to the distribution of the perturbations so long as they are independent across players, have positive support, and are stationary. In addition, we showed that for 2×2 games the stochastically stable equilibria are independent of m and k so long as m/k and k are sufficiently large. It is not clear whether this result holds for weakly acyclic games in general, although we know of no examples in which it fails to hold.

A second question is whether the stochastically stable equilibria can be characterized more succinctly. The algorithm described in the Appendix shows how to compute the stochastically stable equilibria in a wide class of dynamical models with perturbations, but it does not describe these equilibria in terms of a simple formula. There is no reason to think that such a formula exists in the general case. In specific classes of games, however, one can sometimes exploit the payoff structure to obtain more specific answers. Kandori and Rob (1992) explore this issue for pure coordination games and differentiated-product oligopoly games.

Another class of games where one can obtain an explicit formula for the stochastically stable equilibria is the bargaining problem. Consider the two-person Nash demand game in which each player demands a share of a fixed pie. They get their shares if the shares sum to 1 or less; otherwise they get nothing. The strategy spaces can be made discrete by assuming that the shares are rounded to a large, fixed number of decimal places. Let two disjoint populations of agents play this game adaptively, where all members of one population have utility function u and the others have utility function v . Then the stochastically stable equilibria are close to the Nash solution. (See Young (1993).)

A third issue is whether and how the stochastically stable equilibria change when the agents are allowed more decision-making scope. For example, what happens when the agents learn as they play the game over time, or make inferences about the others' decision rules, or choose optimal sample sizes based on their costs of gathering information? These additions complicate both the state space and the stochastic process. They may also require more common knowledge on the part of the agents. In any event, if the agents make mistakes infrequently and independently of each other, then the stochastically stable states can still be analyzed using the general techniques developed here. We have deliberately chosen to focus on the case where agents do *not* learn in

order to show that convergence to equilibrium can occur with no common knowledge and with only a minimum degree of rationality on the part of the agents. Society can “learn” even when its members do not.

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APPENDIX

Here we shall prove a general result on finite Markov chains of which Theorem 2 is a special case. This result amounts to a finite version of results obtained by Freidlin and Wentzell for continuous diffusion processes (see Freidlin and Wentzell (1984, pp. 186–187)). Let P^0 be a stationary Markov chain defined on a finite state space X . Suppose that this process is subjected to a small perturbation or noise. By this we mean that with high probability the process follows the transition function P^0 , but with small probability certain transitions occur that could *not* have occurred via P^0 . We shall assume that the perturbed process can be modelled as a stationary Markov chain on X with transition function P^ε , where ε is a scalar parameter that measures the overall level of noise, ε takes on all values in some interval $(0, a]$, and the following conditions hold for all $x, y \in X$:

(6) P^ε is aperiodic and irreducible for all $\varepsilon \in (0, a]$,

(7) $\lim_{\varepsilon \rightarrow 0} P_{xy}^\varepsilon = P_{xy}^0$,

(8) $P_{xy}^\varepsilon > 0$ for some ε implies $\exists r \geq 0$ s.t. $0 < \lim_{\varepsilon \rightarrow 0} \varepsilon^{-r} P_{xy}^\varepsilon < \infty$.

Condition (6) implies that the perturbed process has a unique stationary distribution μ^ε for every $\varepsilon \in (0, a]$. Condition (7) says that the perturbed process converges to the unperturbed one. Condition (8) says that the transition $x \rightarrow y$ is either impossible in the perturbed process P^ε for all $\varepsilon \in (0, a]$, or P_{xy}^ε is of order ε^r for some unique real number $r \geq 0$ as ε becomes small. In the latter case we set $r(x, y) = r$ and call $r(x, y)$ the *resistance* of the transition $x \rightarrow y$. By virtue of (7), $r(x, y) = 0$ if and only if $P_{xy}^0 > 0$. Thus the transitions of zero resistance are the same as the feasible transitions under P^0 . Any family of Markov processes P^ε satisfying (6)–(8) will be called a *regular perturbation* of P^0 .

The family P^ε defined by (2) in the text is a regular perturbation of the process P^ε defined in (1), and the resistance of a one-period transition is the minimum number of mistakes required to make it.

By hypothesis, the perturbed process P^ε is aperiodic and irreducible, so it has a unique stationary distribution μ^ε for every $\varepsilon > 0$. The unperturbed process, by contrast, may have many stationary distributions. We are going to show that $\lim_{\varepsilon \rightarrow 0} \mu^\varepsilon = \mu^0$, where μ^0 is one of the stationary distributions of P^0 . Thus the perturbations effectively *select* among the stationary distributions of P^0 . The support of the stationary distribution μ^0 is a subset of the recurrent communication classes of P^0 . Thus the perturbations effectively select among the recurrent communication classes. In fact, the perturbations typically select *exactly one* communication class of P^0 . The selected class (or classes) are computed by finding a path of least resistance from every recurrent communication class to every other. Hence the selected classes depend only on the resistances $r(x, y)$, that is, only on the *order of magnitude* of the various perturbations. This is important in applications, where the general form of the perturbations may be known, but not their precise values.

To characterize the limiting distribution μ^0 , we shall define two directed graphs. The first graph G has vertex set X (the set of all states) and there is a directed edge from state x to state y if and only if the one-period transition $x \rightarrow y$ has positive probability under P^ε for all sufficiently small $\varepsilon > 0$. In this event, let $r(x, y)$, as defined by (8), be the weight or *resistance* of the directed edge (x, y) .

Let the recurrent communication classes of P^0 be denoted by $X_1, \dots, X_J$. These classes can be characterized within the graph G as follows: (i) from every vertex there exists a path of zero

resistance to at least one of the classes X_i ; (ii) for every two vertices x and y within the same class X_i there is a path of zero resistance from x to y and vice versa; (iii) every edge from a vertex in X_i to a vertex not in X_i has positive resistance.

For every $i \neq j$ let r_{ij} be the *least resistance* among all directed paths that begin in X_i and end in X_j . (For this purpose it is sufficient to fix any two states $x \in X_i$ and $y \in X_j$ and find a least-resistant path from x to y .) This is well-defined because there is at least one path from every class to every other, by virtue of the assumption that P^ε is irreducible when $\varepsilon > 0$.

Now define a second graph $\mathcal{S}$ as follows: the vertices are the indices $\{1, 2, \dots, J\}$, and for each pair (i, j) there is a directed edge from i to j with weight r_{ij} . $\mathcal{S}$ is normally much smaller than G ; in fact it may have only a few vertices. In the case of a 2×2 coordination game, for example, adaptive play has only two recurrent classes (assuming $k \leq m/3$), so $\mathcal{S}$ has exactly two vertices.

Define a *j-tree* in $\mathcal{S}$ to be a spanning subtree of $\mathcal{S}$ such that for every vertex $i \neq j$ there exists exactly one directed path from i to j . Let $\mathcal{T}_j$ be the set of all *j-trees* in $\mathcal{S}$. For each j find a *j-tree* of least total resistance, and let this resistance be denoted by γ_j . γ_j is the *stochastic potential* of the class X_j .

We shall now prove the following result, of which Theorem 2 is a special case.

THEOREM 4: *Let P^0 be a stationary Markov process on the finite state space X with recurrent communication classes $X_1, \dots, X_J$. Let P^ε be a regular perturbation of P^0 , and let μ^ε be its unique stationary distribution for every small positive ε . Then:*

- (i) *as $\varepsilon \rightarrow 0$, μ^ε converges to a stationary distribution μ^0 of P^0 .*
- (ii) *x is stochastically stable ($\mu_x^0 > 0$) if and only if x is contained in a recurrent class X_j that minimizes γ_j .*

Freidlin and Wentzell (1984, pp. 186–187) prove an analogous result when the unperturbed process is a deterministic dynamical system described by a differential equation on a manifold, and the perturbed process is a family of diffusion processes whose drift converges to that of the unperturbed process as the diffusion goes to zero. The minimum resistance between two states is found by integrating a certain “action functional” along all continuous paths from one state to the other and taking the infimum over all such paths. (See Freidlin and Wentzell (1984, p. 161).) However, their result requires numerous regularity conditions not needed here (Freidlin and Wentzell (pp. 155 and 169)), and in any event the characterization given in Theorem 4 is much simpler analytically.

The proof will be divided into two lemmas. In the first, we establish statement (i) and show how to characterize the stochastically stable states by solving a series of arborescence problems in the graph G . In the second lemma we show that this characterization can be reduced to the simpler problem of solving arborescence problems in the (typically) much smaller graph $\mathcal{S}$.

We begin by characterizing the stochastically stable states in terms of the graph G , whose vertex set is the whole state space X . For any vertex z of G a *z-tree* T is a spanning tree in G such that, for every vertex $x \neq z$, there exists a unique directed path from x to z . Let $\mathcal{T}_z$ be the set of all *z-trees* in G and define

$$(9) \quad \gamma(z) = \min_{T \in \mathcal{T}_z} \sum_{[x, y] \in T} r(x, y).$$

The following result generalizes Theorem 1 in Kandori, Mailath, and Rob (1993), and utilizes a lemma on Markov chains due to Freidlin and Wentzell.⁸

LEMMA 1: *Let P^ε be a regular perturbation of P^0 and let μ^ε be its stationary distribution. Then $\lim_{\varepsilon \rightarrow 0} \mu^\varepsilon = \mu^0$ exists and μ^0 is a stationary distribution of P^0 . Moreover, $\mu_x^0 > 0$ if and only if $\gamma(x) \leq \gamma(y)$ for all y in X .*

⁸ Using different methods, Canning (1992) proves that μ^ε converges to a stationary distribution μ^0 of P^0 under more general conditions, but he does not characterize the support of the limiting distribution.

PROOF: Let P' be the transition function of any aperiodic, irreducible, stationary Markov process defined on the finite space X . For each $z \in X$, define the number

$$p'_z = \sum_{T \in \mathcal{T}_z} \prod_{(x,y) \in T} P'_{xy},$$

where p'_z is positive because P' is irreducible. Let

$$\mu'_z = p'_z / \left(\sum_{x \in X} p'_x \right) > 0.$$

It may then be verified that $\mu' P' = \mu'$, from which it follows that μ' is the unique stationary distribution of P' . (See Freidlin and Wentzell (1984, Chapter 6, Lemma 3.1).)

Now let us apply this result to the process P^ε hypothesized in Lemma 1. Let

$$(10) \quad p_z^\varepsilon = \sum_{T \in \mathcal{T}_z} \prod_{(x,y) \in T} P_{xy}^\varepsilon.$$

By the above result, the stationary distribution of P^ε is given by the formula

$$(11) \quad \mu_z^\varepsilon = p_z^\varepsilon / \sum_{x \in X} p_x^\varepsilon.$$

Define $\gamma(z)$ as in (9) and let $\gamma^* = \min_z \gamma(z)$. We are going to show that $\mu_z^\varepsilon > 0$ if and only if $\gamma(z) = \gamma^*$. Choose a z -tree T with resistance $\gamma(z)$ and consider the identity

$$(12) \quad \varepsilon^{-\gamma^*} \prod_{(x,y) \in T} P_{xy}^\varepsilon = \varepsilon^{r(T)-\gamma^*} \prod_{(x,y) \in T} \varepsilon^{-r(x,y)} P_{xy}^\varepsilon.$$

By (8),

$$(13) \quad \lim_{\varepsilon \rightarrow 0} \varepsilon^{-r(x,y)} P_{xy}^\varepsilon > 0 \quad \text{for every } (x,y) \in T.$$

If $r(T) = \gamma(z) > \gamma^*$, it follows from (12) and (13) that

$$\lim_{\varepsilon \rightarrow 0} \varepsilon^{-\gamma^*} \prod_{(x,y) \in T} P_{xy}^\varepsilon = 0,$$

so

$$\lim_{\varepsilon \rightarrow 0} \varepsilon^{-\gamma^*} p_z^\varepsilon = 0.$$

Similarly, if $r(T) = \gamma(z) = \gamma^*$ we obtain

$$(14) \quad \lim_{\varepsilon \rightarrow 0} \varepsilon^{-\gamma^*} p_z^\varepsilon > 0.$$

From (13), (14), and the identity

$$\mu_z^\varepsilon = \varepsilon^{-\gamma^*} p_z^\varepsilon / \sum_{x \in X} \varepsilon^{-\gamma^*} p_x^\varepsilon,$$

it follows that

$$\lim_{\varepsilon \rightarrow 0} \mu_z^\varepsilon = 0 \quad \text{if } \gamma(z) > \gamma^*$$

and

$$\lim_{\varepsilon \rightarrow 0} \mu_z^\varepsilon > 0 \quad \text{if } \gamma(z) = \gamma^*.$$

Thus we have shown that $\lim_{\varepsilon \rightarrow 0} \mu^\varepsilon$ exists, and its support is the set of states z that minimize $\gamma(z)$. Since μ^ε satisfies the equation $\mu^\varepsilon P^\varepsilon = \mu^\varepsilon$ for every $\varepsilon > 0$, it follows from assumption (7) that $\mu^0 P^0 = \mu^0$. Hence μ^0 is a stationary distribution of P^0 . This completes the proof of Lemma 1.

Since μ^0 is a stationary distribution of P^0 , $\mu_z^0 = 0$ for every state z that is not recurrent under P^0 . To find the stochastically stable states, it therefore suffices to compute the potential function only on the recurrent states.

It is easy to see that all states in the same recurrent communication class have the same potential. Suppose, in fact, that x is in X_j and T is an x -tree in G with potential $\gamma(x)$. Let y be some other vertex in X_j . Choose a path of zero resistance from x to y . The union of this path and T contains a y -tree T' that has the same resistance as T . From this it follows that the potential of y is no larger than the potential of x , and a symmetric argument shows that it is no smaller. Hence they have the same potential.

We shall now show that the potential on each recurrent class X_j is precisely γ_j , namely, the least resistance among all j -trees in the graph $\mathcal{G}$. Thus the potential may be computed simply by solving an arborescence problem on the reduced graph $\mathcal{S}$.

LEMMA 2: $\gamma(x) = \gamma_j$ is the stochastic potential of all states $x \in X_j$.

PROOF: Fix one state x_j in each recurrent class X_j . We shall show first that $\gamma(x_j) \leq \gamma_j$; then we shall show the reverse inequality.

Fix a class X_j and a j -tree τ in $\mathcal{S}$ whose resistance $r(\tau)$ equals γ_j . For every $i \neq j$, there exists exactly one outgoing edge $(i, i') \in \tau$. In the graph G (whose vertices are the states) choose a directed path $D_{ii'}$ from x_i to $x_{i'}$ having resistance $r_{ii'}$. Now choose a directed subtree T_i that spans the vertex set X_i such that from every vertex in X_i there is a unique directed path to $x_{i'}$. Since X_i is a communication class of P^0 , T_i can be chosen so that it has zero resistance.

Let E be the union of all of the edges in the trees T_i and all of the edges in the directed paths $D_{ii'}$ where $(i, i') \in \tau$. By construction, E contains at least one directed path from every vertex in X to the fixed vertex x_j . Therefore it contains a subset of edges that form an x_j -tree T in G . The resistance of T is clearly less than or equal to the sum of the resistances of the paths $D_{ii'}$, so it is less than or equal to $r(\tau)$. Thus $\gamma(x_j) \leq \gamma_j = r(\tau)$ as claimed.

To show that $\gamma(x_j) \geq \gamma_j$, fix a class j and a least-resistant x_j -tree T among all x_j -trees on the vertex set X . Label each of the specially chosen vertices x_i by the class " i " to which it belongs. These will be called *special vertices*. A *junction* in T is any vertex y with at least two incoming T -edges. If the junction y is not a special vertex, label it " i " if there exists a path of zero resistance from y to X_i . (There exists at least one such class because they are the recurrent classes of P^0 ; if there are several such classes choose any one of them as label.) Every labelled vertex is either a special vertex or a junction (or both), and the label identifies a class to which there is a path of zero resistance.

Define the *special predecessors* of a state $x \in X$ to be the special vertices x_i that strictly precede x in the fixed tree T (i.e., such that there is a path from x_i to x in T) and such that there is no other special vertex x_j on the path from x_i to x .

(15) *If x_i is a special predecessor of a labelled vertex x , then the unique path in the tree from x_i to x has resistance at least r_{ik} , where k is the label of x .*

Property (15) clearly holds for the tree T because any path from the special vertex x_i to a vertex labelled " k " can be extended by a zero resistance path to the class X_k , and the total path must have resistance at least r_{ik} . We shall now perform certain operations on the tree T that preserve property (15), and that bring it into a form that is more or less congruent to a j -tree in $\mathcal{S}$. We shall do this by successively eliminating all junctions that are not special vertices.

Suppose that T contains a junction y that is not a special vertex, and let its label be " k ". We distinguish two cases, depending on whether the special vertex x_k is or is not a predecessor of y in the tree.

Case 1: If x_k is not a predecessor of y in the tree (see Figure 4), cut off the subtree consisting of all edges and vertices that precede y and glue them onto the tree at the vertex x_k .

Case 2: If x_k is a predecessor of y (see Figure 5), cut off the subtree consisting of all edges and vertices that precede y (except for the path from x_k to y and all of its predecessors) and glue them onto x_k .

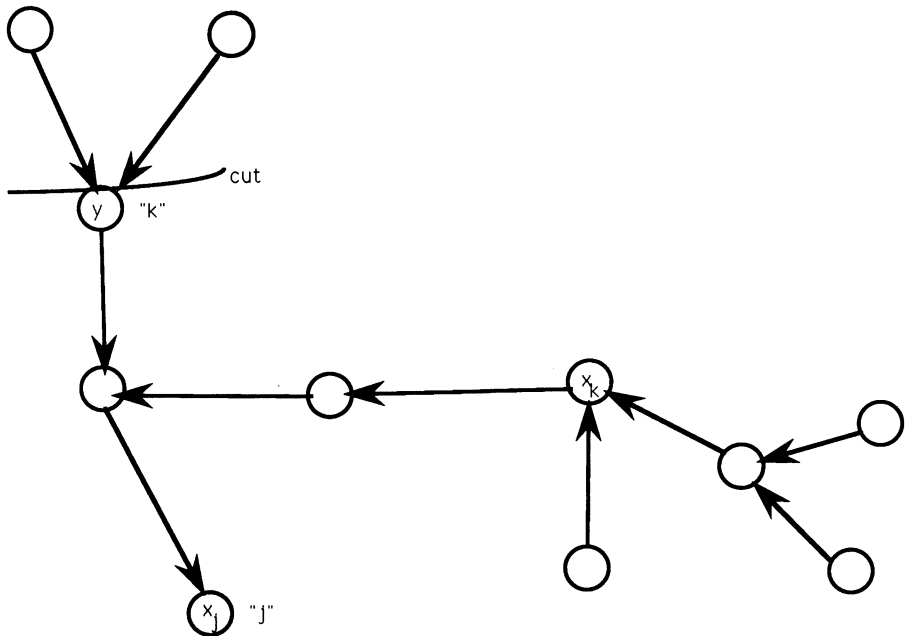


FIGURE 4a.—Case 1 surgery: before.

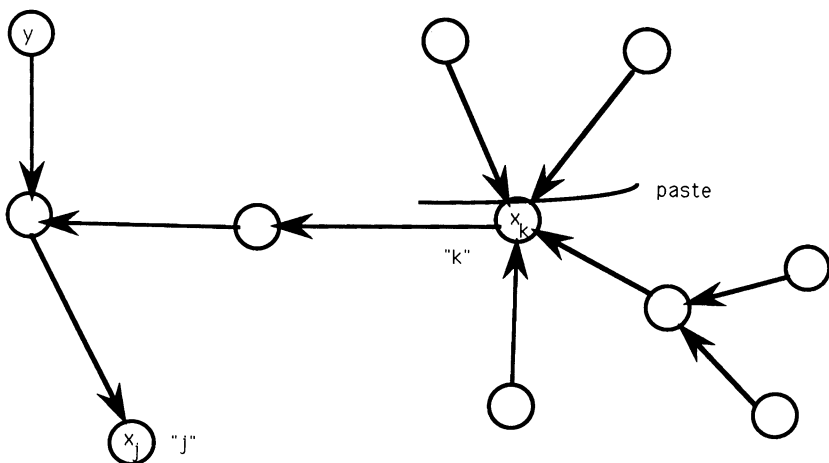


FIGURE 4b.—Case 2 surgery: after.

Both of these operations preserve property (15) because x_k and y have the same label. Each operation reduces by one the number of junctions that are not special vertices. Thus we eventually obtain an x_j -tree T^* in which every junction is a special vertex, property (15) is satisfied, and T^* has the same resistance as the original tree T .

Now construct a j -tree τ on the vertex set J as follows. For every two classes i and i' put the directed edge (i, i') in τ if and only if x_i is a special predecessor of $x_{i'}$ in T^* . By construction, τ forms a j -tree. Let $D_{ii'}^*$ be the unique path in T^* from x_i to $x_{i'}$. By property (15) its resistance is at

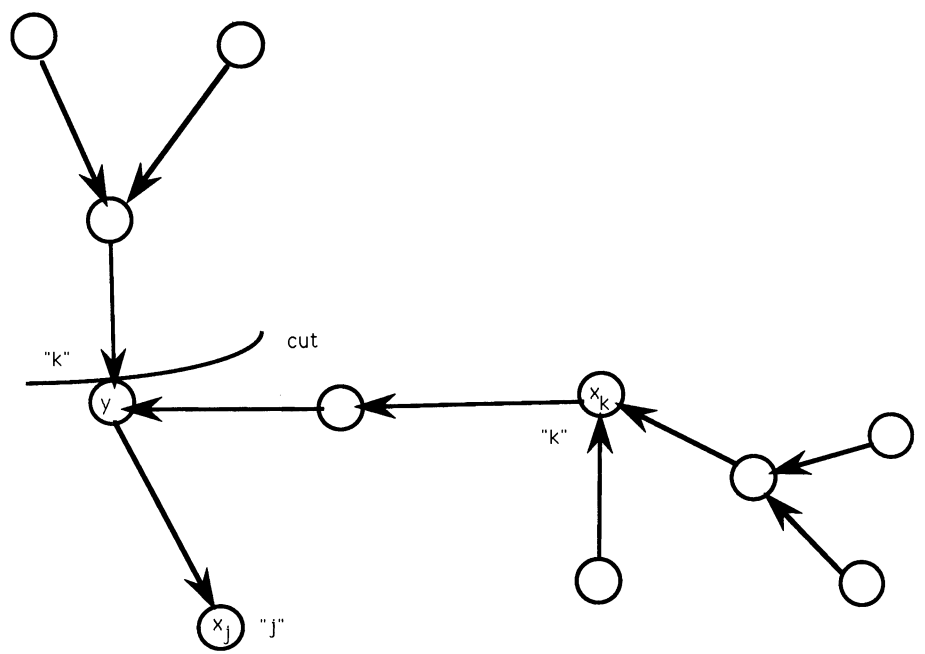


FIGURE 5a.—Case 2 surgery: before.

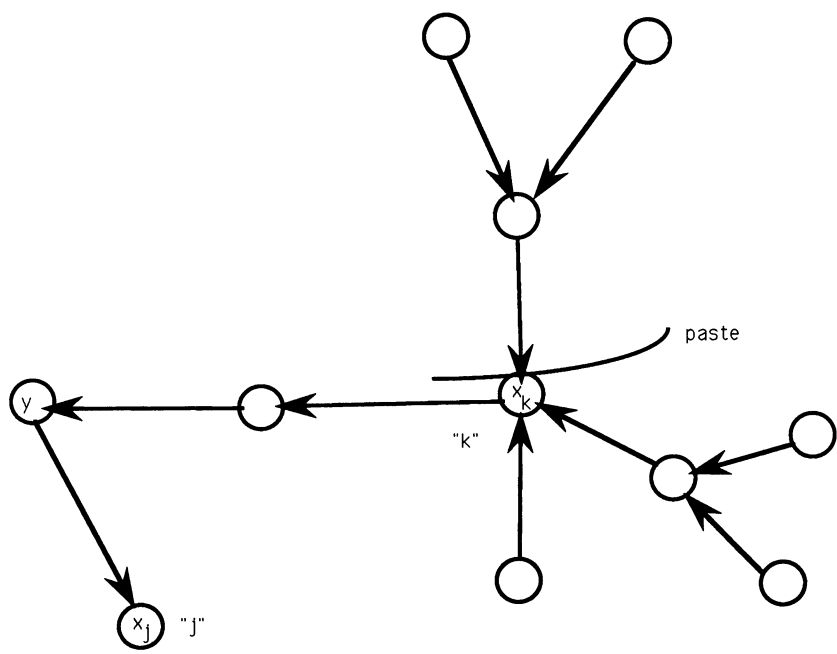


FIGURE 5b.—Case 2 surgery: after.

least $r_{ii'}$. The paths $D_{ii'}^*$ are edge-disjoint because every junction is one of the special vertices. Since T^* contains their union, the resistance of T^* is at least $\sum_{(i,i') \in \tau} r_{ii'}$. But $\sum_{(i,i') \in \tau} r_{ii'}$ is the resistance of τ . Hence $\gamma(x_i) = r(T^*) \geq \sum_{(i,i') \in \tau} r_{ii'} = r(\tau) \geq \gamma_j$ as claimed.

This completes the proof of Lemma 2, which, together with Lemma 1, establishes Theorem 4.

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